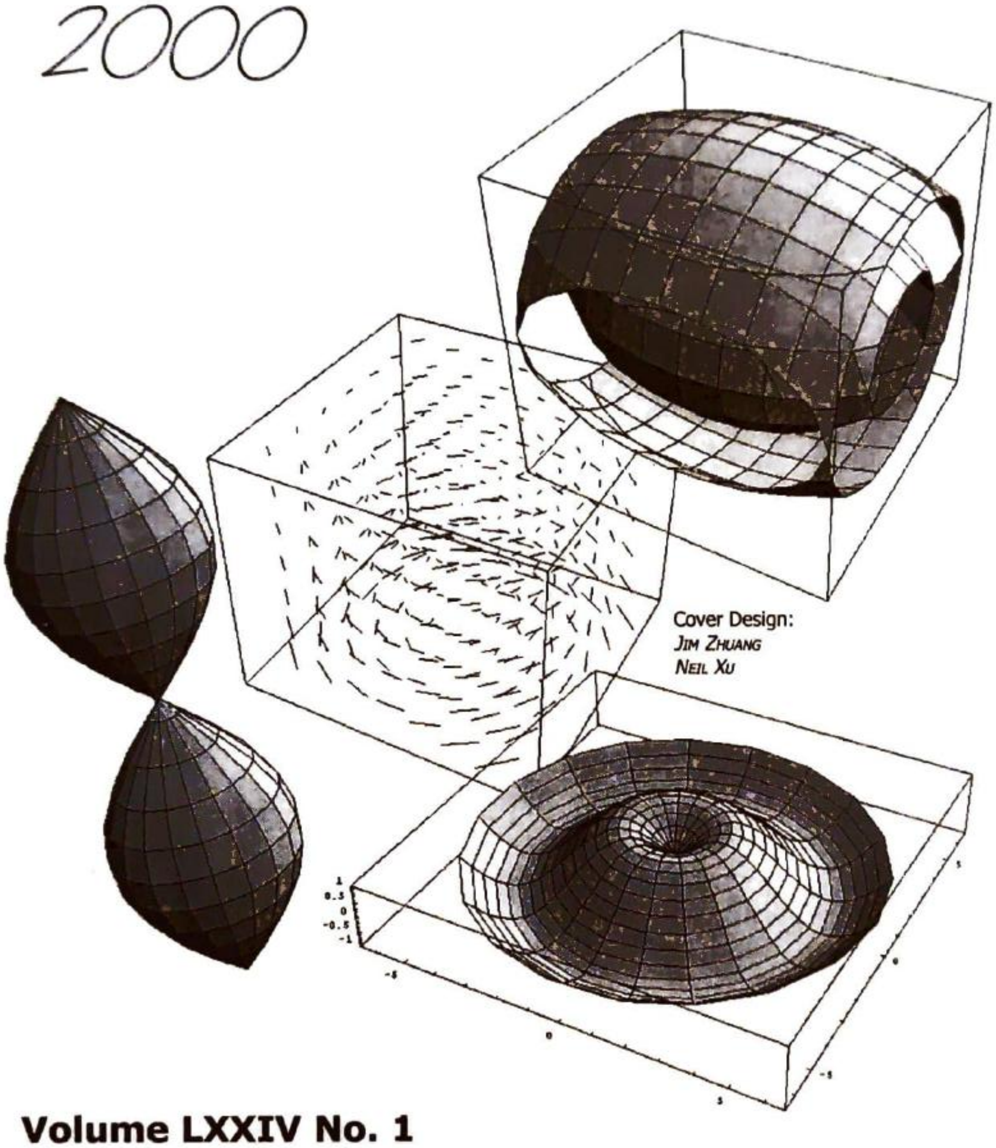
MATH SURVEY

2000



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Mathematical Operators and Their Applications in Physics

Jerry Moy Chow

Introduction:

The homogeneous second order linear partial differential equation is a typical form of many interesting problems in physics and in mathematics. The separation of variables will lead to second order ordinary differential equations in such a form of an eigenvalue equation of a differential operator as

$$\left[\frac{d^2}{dx^2}+\left(\lambda-V(x)\right)\right]\psi=0 \qquad (1)$$

which is known as the one dimensional Schrödinger equation with a coordinate dependence of potential energy V(x). Solutions of equation (1) for different V(x) can be found in textbooks of mathematics. However, the calculation is very tedious. In elementary algebra, a quadratic equation is factorized into two linear equations and then it is easily solved by solving two linear equations. Following the technique of factorization in elementary algebra, we are going to factorize the second order differential operator into two first order ones so that a simple calculation of operator algebra in abstract vector space will easily provide solutions to the Shcrödinger equation. It is interesting that the product of the factorized operators does not give back the original operator but with an addition of a commutator which is negligible for classical mechanics but important for quantum mechanics. Therefore, the new algebra for the calculation of factorized operators will provide an interesting and powerful method for solving problems in atomic and subatomic physics. The particle-like behaviors of the operators suggest further quantization of the wave, as it is demonstrated by the successful application to black body radiation.

Factorization of Operators and the Eigenvalues and Eigenfunctions:

In order to show how such a technique works, let us consider as an example $V(x) = \beta x^2$, and $\lambda = \alpha E$. When $\beta = (m^2\omega^2)/\hbar^2$ and $\alpha=(2M)/\hbar^2$, it corresponds to a harmonic oscillator of mass M, and energy $\hbar\omega$. Equations (1) will become

$$\left(--\frac{\hbar^2}{2M}\frac{d^2}{dx^2}+\frac{1}{2}M\omega^2x^2\right)\psi=E\psi \qquad (2)$$

After a substitution of
into equations (2), we obtain

$$\left(-\frac{d^2}{dQ^2}+Q^2\right)\frac{\hbar\omega}{2}\psi = E\psi \qquad (3).$$

$$x=\sqrt{\frac{\hbar}{M\omega}}Q$$

$$L=\frac{\hbar\omega}{2}\left(i^2\frac{d^2}{dQ^2}+Q^2\right)=\sqrt{\frac{\hbar\omega}{2}}\left(Q+i\,i\frac{d}{dQ}\right)\cdot\sqrt{\frac{\hbar\omega}{2}}\left(Q-i\,i\frac{d}{dQ}\right)$$

The operator in (3) can be factorized as
which gives the following two first order differential operators,

$$R=\sqrt{\frac{\hbar\omega}{2}}\left(Q-i\,i\frac{d}{dQ}\right)=\sqrt{\frac{\hbar\omega}{2}}\left(Q+\frac{d}{dQ}\right) \qquad (4)$$

and

one being the adjoint of the other.

$$R^+=\sqrt{\frac{\hbar\omega}{2}}\left(Q+i\,i\frac{d}{dQ}\right)=\sqrt{\frac{\hbar\omega}{2}}\left(Q-\frac{d}{dQ}\right), \qquad (5)$$

Theorem 1. Operators $\frac{d}{dQ}$ and Q are not commute, and the commutation rule is $\left[\frac{d}{dQ},Q\right]=1$.

Proof : $\left(Q\frac{d}{dQ}\right)\psi=Q\frac{d\psi}{dQ}$

$$\left(\frac{d}{dQ}Q\right)\psi=\frac{d}{dQ}(Q\psi)=\psi+Q\frac{d\psi}{dQ}=\psi+\left(Q\frac{d}{dQ}\right)\psi$$

$$\left[\left(\frac{d}{dQ}Q\right)-\left(Q\frac{d}{dQ}\right)\right]\psi=\psi$$

$$\therefore \quad \frac{d}{dQ}Q-Q\frac{d}{dQ}=\left[\frac{d}{dQ},Q\right]=1$$

Q.E.D.

Theorem 3. Operators R and R^+ are not commute, and the commutation rule is $\left[R,R^+\right]=\hbar\omega$

Proof : Following a similar procedure as in the proof of theorem 2, it can be shown that

$$R^+R=L-\frac{\hbar\omega}{2} \qquad (7)$$

From Theorem 2 $\quad RR^+=L+\frac{\hbar\omega}{2}$

$$\therefore \qquad RR^+-R^+R=\left[R,R^+\right]=\hbar\omega \qquad (8)$$

Q.E.D.

Since $R^+R|\varepsilon_{min}\rangle = 0 = (L - \frac{1}{2}\hbar\omega)|\varepsilon_{min}\rangle = (\varepsilon_{min} - \frac{1}{2}\hbar\omega)|\varepsilon_{min}\rangle = 0,$

we have $\varepsilon_{min} - \frac{1}{2}\hbar\omega = 0$, or $\varepsilon_{min} = \frac{1}{2}\hbar\omega$, and $|\varepsilon_{min}\rangle = |0\rangle$,

the zero point ground state energy corresponding to the state of vacuum $|0\rangle$.

The results are

$$L|0\rangle = \frac{1}{2}\hbar\omega|0\rangle$$

and

$$L|n\rangle = \left(n + \frac{1}{2}\right)\hbar\omega|n\rangle \tag{14}$$

From Theorem 1, $\left[\frac{d}{dQ}, Q\right] = 1$

$$\therefore \; RR^+ = L + \frac{\hbar\omega}{2} \tag{6}$$

Q.E.D.

Theorem 2. The factorization of the operator L gives two operators R and R^+, but the product of R and R^+ does not give L but $L + \frac{1}{2}\hbar\omega$.

Proof : $RR^+ = \sqrt{\frac{\hbar\omega}{2}}\left(Q + \frac{d}{dQ}\right)\sqrt{\frac{\hbar\omega}{2}}\left(Q - \frac{d}{dQ}\right) = \frac{\hbar\omega}{2}\left[Q\left(Q - \frac{d}{dQ}\right) + \frac{d}{dQ}\left(Q - \frac{d}{dQ}\right)\right]$

$$RR^+ = \frac{\hbar\omega}{2}\left[Q^2 - Q\frac{d}{dQ} + \frac{d}{dQ}Q - \frac{d^2}{dQ^2}\right] = \frac{\hbar\omega}{2}\left[-\frac{d^2}{dQ^2} + Q^2\right] + \frac{\hbar\omega}{2}\left[\frac{d}{dQ}, Q\right]$$

Theorem 4. Operators R^+ and R are operators with step - up and step - down eigenvalues respectively.

Proof : Let ε and $|\varepsilon\rangle$ be the eigenvalue and eigenfunction of the operator L, that is

$$L|\varepsilon\rangle = \varepsilon|\varepsilon\rangle$$

then

$$LR^+|\varepsilon\rangle = (R^+L + \hbar\omega R^+)|\varepsilon\rangle = (\varepsilon + \hbar\omega)R^+|\varepsilon\rangle,$$

$$LR^+R^+|\varepsilon\rangle = (R^+L + \hbar\omega R^+)R^+|\varepsilon\rangle = (\varepsilon + 2\hbar\omega)R^+R^+|\varepsilon\rangle$$

After repeating the process n times, we get

$$L(R^+)^n|\varepsilon\rangle = (\varepsilon + n\hbar\omega)(R^+)^n|\varepsilon\rangle \tag{11}$$

Similarly,

$$L(R)^n|\varepsilon\rangle = (\varepsilon - n\hbar\omega)R^n|\varepsilon\rangle \tag{12}$$

Therefore, R^+ and R give step - up and step - down eigenvalues respectively.

The eigenvalues are energies that cannot be negative. In order to get a minimum

$$L|\varepsilon_{\min}\rangle = \varepsilon_{\min}|\varepsilon_{\min}\rangle, \quad \text{and} \quad LR|\varepsilon_{\min}\rangle = (\varepsilon_{\min} - \hbar\omega)R|\varepsilon_{\min}\rangle = 0,$$

$$\therefore \quad R|\varepsilon_{\min}\rangle = 0 \tag{13}$$

energy eigenvalue, we must set

$$R|0\rangle = \sqrt{\frac{\hbar\omega}{2}}\left(Q + \frac{d}{dQ}\right)|0\rangle = 0,$$

$$\therefore \qquad Q\langle Q|0\rangle + \frac{d}{dQ}\langle Q|0\rangle = 0, \quad \text{or} \quad \frac{d}{dQ}\psi_0(Q) = -Q\psi_0(Q).$$

The following useful commutation rules can be easily obtained :

$$[L, R] = -\hbar\omega R \tag{9}$$

$$[L, R^+] = -\hbar\omega R^+ \tag{10}$$

The next is to find the eigenfunctions using $|n\rangle = (R^+)^n|0\rangle$ *and* $R|0\rangle=0$. Because

The ground state eigenfunction is

$$\psi_0(Q) = Ae^{\frac{-Q^2}{2}} \quad \text{or} \quad \psi_0(x) = Ae^{-\frac{M\omega}{2\hbar}x^2} \tag{15}$$

where A is the normalization constant.
The eigenfunctions for excited state are

$$|n\rangle = \frac{1}{\sqrt{n!}}(a^+)^n|0\rangle$$

or

$$\langle x|n\rangle = \psi_n(x) = \frac{1}{\sqrt{n!}}\left(-\sqrt{\frac{\hbar}{M\omega}}\frac{d}{dx} + \sqrt{\frac{M\omega}{\hbar}}x\right)^n Ae^{-\frac{M\omega}{2\hbar}x^2} \tag{16}$$

Consequently, by using the operator factorization technique we have already obtained solutions to the complicated second order differential equation by working with an easy and simple algebra of the operators.

Particle-like Behaviors of the Operators:

It is convenient to define another set of operators to put the results in another

$$H = L - \frac{1}{2}\hbar\omega \tag{17}$$

form. The ground state eigenvalue of energy of $\frac{1}{2}\hbar\omega$ will be infinite for a system of infinite degrees of freedom such as the black body radiation. The zero point energy is a matter of definition. It is convenient to define the Hamiltonian as

and let $$R = \sqrt{\hbar\omega}a, \tag{18}$$

$$R^{+} = \sqrt{\hbar\omega}a^{+}, \tag{19}$$

and $$N = a^{+}a \tag{20}$$

The commutation rule of a and a^{+} is

$$\left[a, a^{+}\right] = \frac{1}{\hbar\omega}\left[R, R^{+}\right] = 1 \tag{21}$$

The operator H can be expressed in terms of a, and a^{+} as follows:

$$H = L - \frac{1}{2}\hbar\omega = RR^{+} - \hbar\omega = \left(aa^{+} - 1\right)\hbar\omega$$

$\because$ $$\left[a, a^{+}\right] = aa^{+} - a^{+}a = 1$$

$\therefore$ $$H = a^{+}a\hbar\omega = N\hbar\omega \tag{22}$$

The eigenvalue equation for operator H can be found as follows:

$\because$ $$L|n\rangle = \left(n + \frac{1}{2}\right)\hbar\omega|n\rangle$$

$\therefore$ $$H|n\rangle = \left(L - \frac{1}{2}\hbar\omega\right)|n\rangle = n\hbar\omega|n\rangle \tag{23}$$

The eigenvalue equation for N is

$$N\hbar\omega|n\rangle = H|n\rangle = n\hbar\omega|n\rangle$$

$\therefore$ $$N|n\rangle = a^{+}a|n\rangle = n|n\rangle \tag{24}$$

Using the relations of $(a^{+})^{+}=a$, $(a)^{+}=a^{+}$, $(|n\rangle)^{*}=(\langle n|)$, $\langle n|n\rangle=(|n\rangle)^{*}(|n\rangle)=1$, we have

$$\left(a|n\rangle\right)^{2} = \left(a|n\rangle\right)^{*}\left(a|n\rangle\right) = \langle n|a^{+}a|n\rangle = \langle n|n\rangle n = n$$

or $$a|n\rangle = \sqrt{n}|n-1\rangle \tag{25}$$

Similarly $$\left(a^{+}|n\rangle\right)^{2} = n+1$$

or $$a^{+}|n\rangle = \sqrt{n+1}|n+1\rangle \tag{26}$$

It agrees with $a|0\rangle = 0$, because $R|0\rangle = \sqrt{\hbar\omega}a|0\rangle = 0$.

Equation (24), (25), and (26) show that operator a^{+} operates on vacuum state $|0\rangle$ and gives $|1\rangle$ state; a operates on $|1\rangle$ and gives vacuum state $|0\rangle$; and $a^{+}a$ operates on $|n\rangle$ state and gives eigenvalue n, an integer in $|n\rangle$ state, leaving the state unchanged. By defining the ground state $|0\rangle$ as the vacuum state and the n^{th} excited state $|n\rangle$ as the n particle or quasi particle state, a, a^{+}, and N act as particle annihilation operator, particle creation operator, and particle number operator respectively. In mathematics writing the operators in the forms of a, a^{+}, and N is for convenience. However, in physics, since a,

a^+, and N have particle-like or quasiparticle-like behaviors, they introduce something new in physics. The discrete quantum state $|n\rangle$ of the harmonic oscillator can be considered as a state occupied by n-particles each of energy $\hbar\omega$ with a total of $n\hbar\omega$. The vacuum state $|0\rangle$ is not occupied by any particle. When operator a operates on a state $|n\rangle$

$$dn_x dn_y dn_z = \frac{l^3}{8\pi^3} dk_x dk_y dk_z = \frac{l^3}{8\pi^3} 4\pi k^2 dk = \frac{l^3}{2\pi^2} k^2 dk$$

with n-particles, it will annihilate one particle to give the $|n\text{-}1\rangle$ state. When operator a^+ operates on a state $|n\rangle$ with n-particle, it will create one particle to give the $|n+1\rangle$ state. On the other hand, when operator N operates on a state $|n\rangle$, it will give the eigenvalue of n while leaving the state unchanged. If the harmonic oscillator represents the sound wave in a lattice vibration, the electromagnetic wave in a cavity in thermal equilibrium, or the wave of the electron in a solid, then the states $|n\rangle$ with n particles or quasiparticles show a further quantization of the wave or the field.

Application to Black-Body Radiation:

It is interesting to apply such an idea to black body radiation. Let us consider in a cavity the electromagnetic radiation as a system of great number of harmonic oscillators in thermal equilibrium at temperature T. Each mode behaves as an independent quantum harmonic oscillator. The state $|n\rangle$ is occupied by n particles named photons, each of energy $\hbar\omega_i$ with an eigenenergy of $n\hbar\omega_i$. Photons are known as Bose particles, the number of which in a state can go from zero to infinity. The probability for a photon to be in a state of energy E is given by the Boltzmann factor $e^{-E/kT}$. Therefore, the average energy of

$$\langle E_i \rangle = \frac{\sum_{n=0}^{\infty} n\hbar\omega e^{-n\hbar\omega/kT}}{\sum_{n=0}^{\infty} e^{-n\hbar\omega/kT}} = \hbar\omega \frac{\sum_{n=0}^{\infty} ne^{-nx}}{\sum_{n=0}^{\infty} e^{-nx}} = \hbar\omega\langle n\rangle \qquad (27)$$

a single mode is

$$\langle n \rangle = \frac{\sum_{n=0}^{\infty} ne^{-nx}}{\sum_{n=0}^{\infty} e^{-nx}} \qquad (28)$$

where $x = \hbar\omega/kT$. The average number of photons is
Let $u = e^{-x}$, then we have

To find the number of modes of oscillations of the electromagnetic wave in the cavity per unit volume between ω_i and $\omega_i + d\omega$, let us consider the cavity to be a cube of side l. Using the periodic condition of the electromagnetic wave on the walls of the cavity, the number of waves in x, y, and z, the n_x, n_y, and n_z are integers, that is,
The number of modes between k and $k + dk$ is

$$\langle n \rangle = \frac{0 + 1u + 2u^2 + 3u^3 + \cdots}{1 + u + u^2 + u^3 + \cdots} = \frac{u(1 + 2u + 3u^2 + \cdots)}{1 + u + u^2 + u^3 + \cdots} = u\frac{d}{du}\ln(1 + u + u^2 + u^3 + \cdots) = u\frac{d}{du}\ln\frac{1}{1-u}$$

$$\langle n \rangle = u \cdot (1-u)\frac{1}{(1-u)^2} = \frac{u}{1-u} = \frac{e^{-\hbar\omega_i/kT}}{1 - e^{-\hbar\omega_i/kT}} = \frac{1}{e^{\hbar\omega_i/kT} - 1} \qquad (29)$$

$$n_x = \frac{l}{\lambda_-} = \frac{k_x l}{2\pi} = \text{integer.}$$

For each frequency (or k) of the electromagnetic wave, there are two polarizations, therefore, the number of modes per unit volume per dk is

$$2\frac{dn_x dn_y dn_z}{l^3} = \frac{k^2}{\pi^2}dk \qquad (30)$$

Because $k=\omega/c$ or $dk=d\omega/c$, the average energy density between ω_i and $\omega_i + d\omega$ is

$$E(\omega)d\omega = \frac{\omega^2 d\omega}{c^3\pi^2}\cdot\frac{\hbar\omega}{e^{\hbar\omega/kT}-1}$$

or

$$E(\nu)d\nu = \frac{8\pi\nu^2}{c^3}\frac{h\nu d\nu}{e^{h\nu/kT}-1} \qquad (31)$$

which is the Planck's Radiation formula. Therefore, the idea of n-particle or quasiparticle state for the n^{th} excited state works successfully.

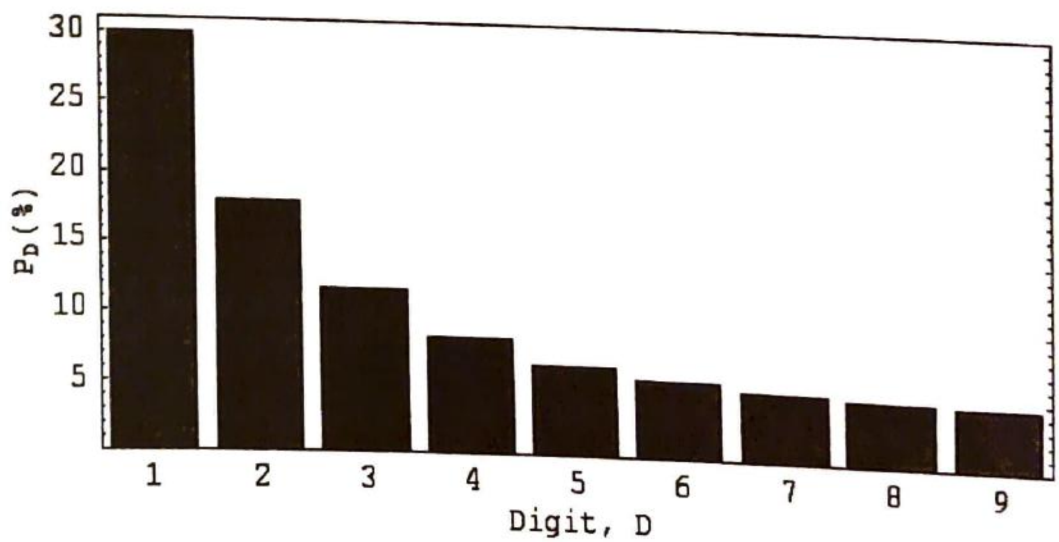
In conclusion, the factorization of higher order differential operator into first order differential operators does provide solutions to the complicated higher order differential equations in a simple and easy way. The product of the linear operators that are factorized from the original operator does not give back the original operator but give back the original operator with the addition of a commutator. The commutator is negligible for classical mechanical system but is extremely important for Quantum Mechanical systems. Therefore, the operator factorization and operator algebra are very important for solving problems in atomic and subatomic problems. Furthermore, the occupation number, creation, and annihilation operators have particle-like or quasiparticle-like behavior. They suggest new ideas to interpret the discrete quantum state $|n\rangle$ as a state of n-particles or n-quasiparticles state, each with energy $\hbar\omega$. Satisfactory application of such an idea to the problem of black body radiation shows that the new idea is successful.

Benford's Law

Daniel Geeng

In 1938 a physicist with the General Electric Company in the U.S., Frank Benford, noticed something strange about a log book he had: the first pages were dirtier than the rest. He suspected that people checked the logs beginning with numeral one more often than the others. After several other observations, he finally reached a surprising conclusions, which today we call Benford's Law.

Benford's law states that for any non-random set of statistical data, the digit 1 will appears as the first digit about 30% of the time. The digit 2 will appear 18% of the time. The trend of decreasing continues all the way to the digit 9, which appears only 4.5% of the time.



To come up with this conclusion, Dr. Benford analyzed data from many different sources. Examples include drainage areas for rivers, numbers appearing in magazine articles, and baseball statistics.

Since it was discovered, mathematicians have tried to some up with a logical explanation. Many different solutions have been offered. Here is one explanation offered by Dr. Mark J. Nigrini, with the use of logic:

"If we think of the Dow Jones stock average as 1000, our first digit would be one. To get a Down Jones average with a first digit of 2, there has to be a 100% increase from 1000. To get an average of 300 from 2000, there needs to be a 66% increase. To get from 3000 to 4000, a 33% increase is needed, As we can clearly see, it is harder to go from 1 to 2 than going from 2 to 3, and so on. Therefore, there should be more numbers beginning with 2 than with 3, and so on."

Other mathematicians have proven properties of Benford's Law. The most significant property was that Benford's law in invariant. This means that if one changes the speed of a car from miles per hour to kilometers per hour, the speed will still follow Benford's Law. The proof of this is credited to Dr. Roger Pinkham of Rutgers University. Benford's Law has also been proven to be base invariant.

The first application of Benford's Law was to detect tax frauds. It had a very high success rate – over 75% - when it was first put to use around 1992. Some people also believe that Benford's law could have been used to detect Y2K problems. Several computer programs based on Benford's Law were used to spot significant changes in accounts between 1999 and 2000, thereby detecting computer problems that might otherwise have gone unnoticed.

Finally, there are some cases when Benford's Law cannot be applied. First, if a list of numbers is completely random, thenthe distribution of the first digits should theoretically be equal. Also, some numbers are special for certain fields and tend to appear more frequently. The probability estimated by Benford's law may not be correct in these cases.

Fractals

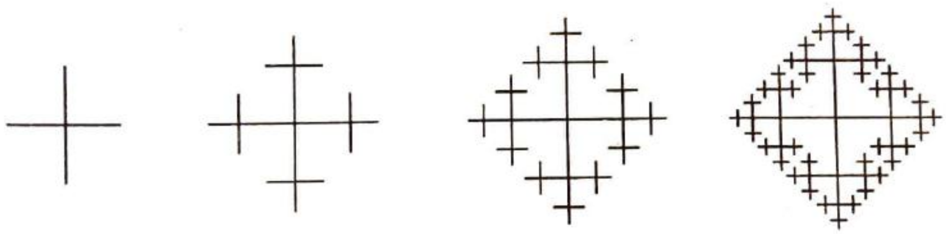
David Cha

What is a fractal anyway? And how is it generated? There is a complex and formal way of defining a fractal. One way is to define fractals as processes or images that exhibit self-similarity. In other words, reduced versions of the fractal appear throughout the fractal. For instance, look at the fractal fern below. Each branch of the fern is a reduced version of the entire fern.

The formal definition involves the concepts of self-similarity, infinite detail, and fractional dimension, all properties of fractals that will be mentioned later.

Fractals are generated by starting with a very simple pattern that grows through the application of rules. In many cases, the rules to make the figure grow from one stage to the next involve taking the original figure and modifying it or adding to it. This process can be repeated recursively an infinite number of times.

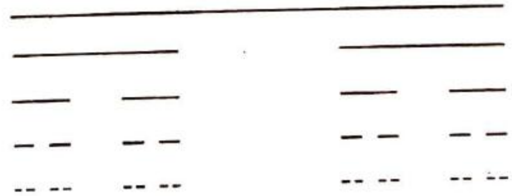
The fractals' growth can be visualized very easily with a simple example below. Start with a + sign and grow it by adding a half size + in each of the four line ends. Repeat the exact same process recursively as many times as desired. This is called the *Plusses* fractal:



The process of creating fractals is called iteration. There are three types of iteration: iterated function system (IFS), generator iteration, and formula iteration. Again, they will all be analyzed later on.

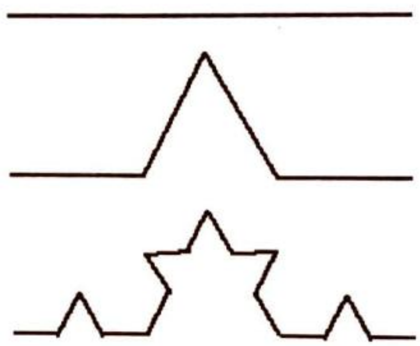
PROPERTIES: We begin with the three properties of fractals. They are self-similarity, infinite detail, and fractional dimension.

Self-similarity is an important feature that exists in virtually every fractal. When a small part of a fractal is magnified, it will look similar to the whole fractal itself. Let us look at the fractal below with strict or perfect self-similarity.



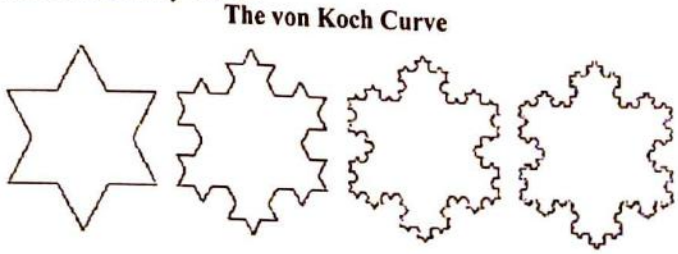
The Cantor Set

The Cantor Set is one of the most famous fractals and the simplest as well. In order to create it, the middle third segment of the first line is removed and we're left with two separate segments. The middle third segments of the resulting lines are removed again. These steps or iterations are repeated infinitely until the Cantor Set is merely a collection of disjointed points. When enlarged, the reduced versions of the image are exact copies of the original and thus strict or perfect self-similarity is present.

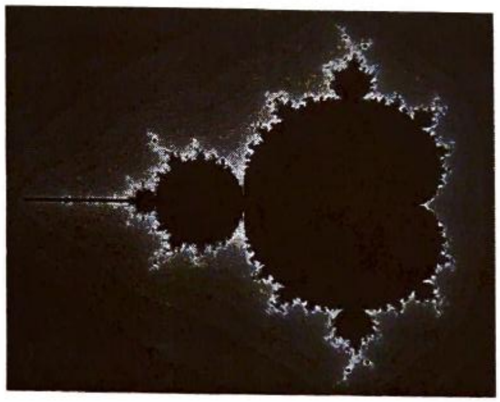


Another perfectly self-similar fractal is the Koch Curve. It is created by dividing a segment into three parts. An equilateral triangle is constructed at center of the segment and the leg that is on the segment is then removed. This iteration is repeated infinitely for all resulting segments. Again, it is evident that perfect self-similarity exists in the Koch Curve.

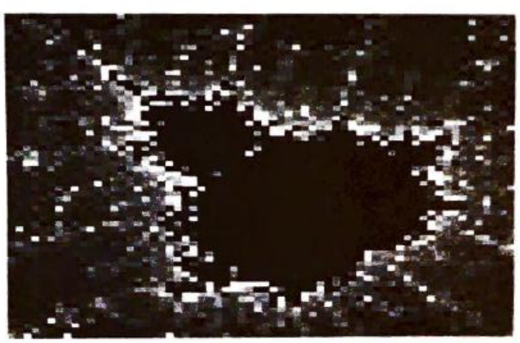
The von Koch Curve



Another type of self-similarity is quasi or approximate-self-similarity. A portion of a set may look similar to the original but will not be identical. For example, in the Mandelbrot Set, one cannot see identical pictures right away. However, when one begins magnifying, he/she will encounter small similar versions of it at all levels of magnification.

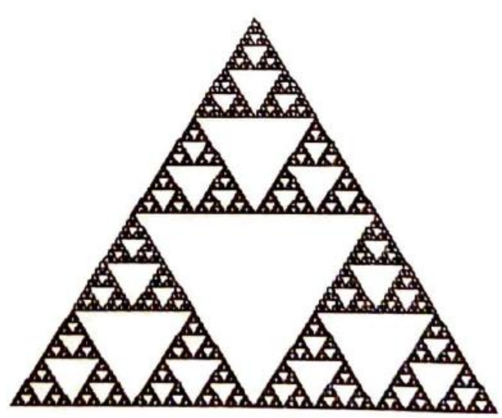


Infinite detail is another common property of fractals. In the fractal below, the Mandelbrot Set, we see that after each magnification, there exists small "bulbs" that look similar to the whole fractal itself. When zooming into the "bulb", there are even smaller bulbs attached to the bigger bulb. No matter how many times one zooms into the picture, there will always a complex image or pattern.

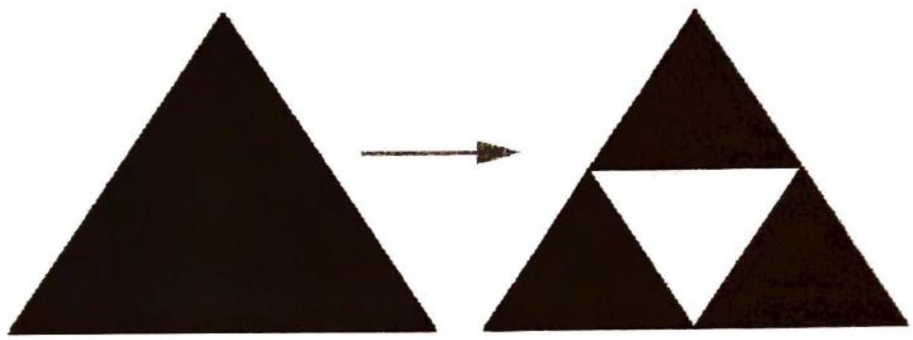


Self-similarity and infinite details can be used to study the shapes and patterns of a fractal. However, they are not exactly defining because they can only help study fractals in a superficial way. But to study fractals in depth, quantifying them will enable one to make comparisons between fractals. The value we assign to a fractal is *fractional dimension*. This is a very complex and hard-to-grasp concept but it revolutionized the field of fractals.

We live in the 3-dimensional world of width, length, and depth. A plane (length and width) is 2-dimensional, a line (length) is 1-dimensional, and a point is 0-dimensional. But take a look at the Koch Curve that was previously shown. One started with a line segment. Then a "kink" was added to the line segment. Next, another "kink" was added to each resulting line segment. After much iteration, the Koch curve is created. Now the kinks cause the line segment to grow towards becoming a plane since it is expanding "height-wise". However, it would be inaccurate to say that the curve is 2-dimensional. It's only 1.2618. That makes sense because it is more than 1-dimensional, but not quite 2-dimensional. The calculation of the fractional dimension will be discussed later on. Another good example is the Sierpinski Triangle, shown below.



To generate the Sierpinski Triangle or Gasket, one starts with an equilateral triangle. It is then divided into 4 smaller equilateral triangles and the center triangle is removed like shown below:



The Sierpinski Triangle starts out with a filled triangle (2-dimensional). Then the middle triangles of the filled triangles are extracted. This is done an infinite number of times. Since one is removing from the triangle, he/she is decreasing the dimension. It first started at 2 and decreased, but will never reach 1(a line). Therefore, the dimension is between 1 and 2. The approximate dimension is 1.5850.

One way to accurately calculate the fractional dimension of a fractal is to employ the similarity method. If a line segment is divided into N equal pieces, there would be many reduced versions of the original line segment, each scaled down by a factor of r. Notice that $Nr = 1$. For a square, divide it into N congruent squares and let the scaling ratio for each side of the square be r. Notice now that the relationship between N and r is $Nr^2 = 1$.

Now we can generalize for d-dimension. One can divide a d-dimensional object into N equal pieces, each having a scaling ratio of r. The relationship will be $Nr^d = 1$.

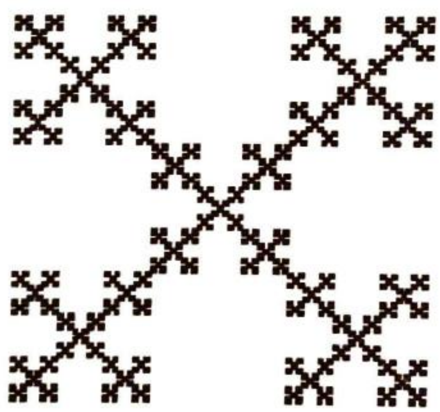
After solving for d algebraically, the equation for fractional dimension is derived to be:

$d = (\log N) / (\log (1/r))$.

(The base of the logarithm can be any positive number other than 1.)

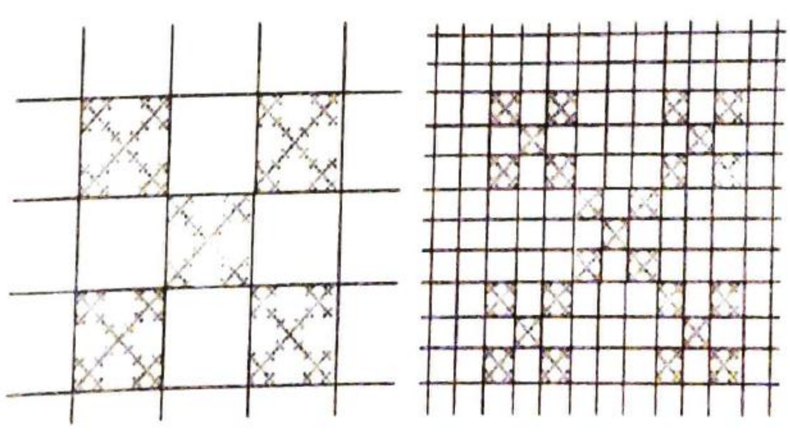
From the Sierpinski Triangle above, one can see that the triangle is divided into 3 equal pieces. The height and width of the triangle is reduced by (1/2). So $N = 3$ and $r = 1/2$.

The equation becomes $d = (\log 3) / (\log 2)$ which is approximately 1.584962500721.



Another method of calculating the fractional dimension is the box counting method. For example, let's calculate the fractal dimension of the Box Fractal shown above.

Using the box-counting method, we put the fractal on a sheet of graph paper. For this fractal, we use boxes with sizes 1/3 and 1/9.



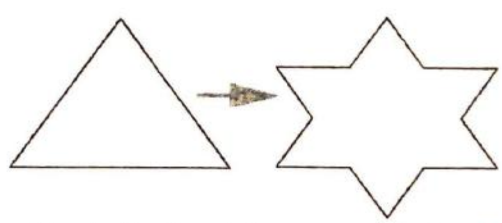
In the first case, 5 boxes are not empty and in the second case, there are 25. Using these numbers, we find that in the first case D = log 5 / log [1/(1/3)] = 1.46. If you do it for the second case, you will find that the answer is the same, which means that our dimension is accurate.

CREATING FRACTALS:

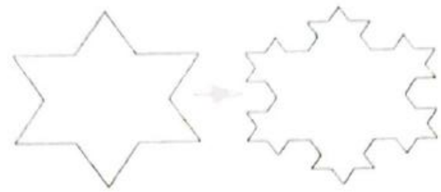
The generation of fractals is done by iteration. Iterating means to repeat a certain series of steps that creates a pattern. There are three types of iterations: generator iteration, iterated function system (IFS), and formula iteration.

Generator iteration is performed by starting with the initial figure called the base and substituting it with the motif.

Take for example, the Koch Curve. It starts with a line segment base and the motif is added to each resulting line segment. The Koch Snowflake below starts with a different base, the triangle, but the same motif is used in this generator iteration.



The next type of iteration is the iterated function system (IFS). A fractal created by this procedure begins with a figure and identical figures are then inserted into the figure. The Sierpinski Gasket, Pyramid, and Carpet are all fractals generated by IFS.



The third and most important type of iteration is formula iteration. Formula iteration has been used to create the most complex fractals. Three types of fractals created by formula iteration are strange attractors, Julia Sets, and the Mandelbrot Set. When viewing a fractal, it should be viewed as a function on a plane made up of many, many points, or pixels. Each pixel has a x-coordinate and a y-coordinate, which determine its position on the plane. Since each pixel is in a different place, each pixel's coordinates are different from the rest. So to start, a pixel is selected. Then a function is iterated on the point to produce new x and y-coordinates. Next the function is iterated on those new coordinates, and so on.

Iteration is continued many, many times until a pattern, the fractal, is finally created. The type of function determines the type of fractal. If a certain type of function is iterated, the Mandelbrot set may be created and if another formula were used, it would yield a Julia set. There are an infinite number of functions, and therefore an infinite number of fractals.

An example of a fractal made by formula iteration is the Henon Attractor.

To create the Henon Attractor the following formulas are used:

New $x = 1 + y - 1.4x^2$

New y = 0.3x

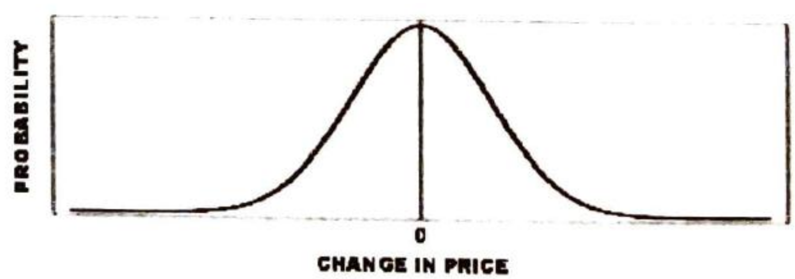
A current point on a plane or in space is used in the formula to calculate the next set of points.

Application

The theory of fractals can be applied to a wide variety of fields, including astronomy, nature and landscape, chemistry, biology, physics, art, music, and most fascinating of all, economics.

In the volatile financial market, fortunes can be made or lost overnight. A firm's stock may drop about 40 percent in one day and rise 30 percent the next. Ever since the start of financial markets, analysts have tried to make predictions for the market but accurate predictions were very difficult to make. For almost a century, investors have adopted the Portfolio Theory, which has been portrayed graphically below.

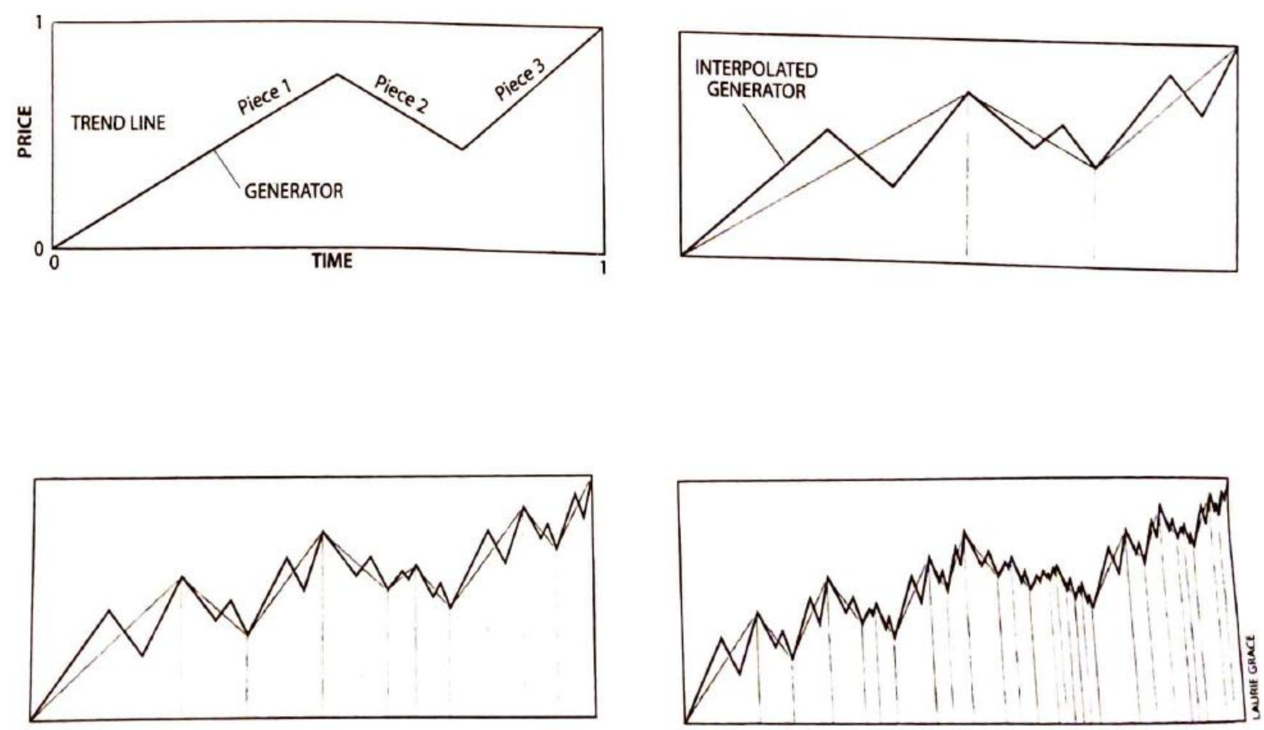
The Portfolio Theory regards large market shifts as unlikely and the theory accounts for about 95 percent of the time. But obviously, the theory does not reflect reality. The remaining 5 percent cannot be ignored because a major event in the market can occur any time. According to the Portfolio Theory, the probability of large fluctuations would be a few millionths of a millionth of a millionth. But great oscillations have occurred on a regular once-a-month basis. This discrepancy precludes exact predictions of future market movements to be made.



Recently, Benoit B. Mandelbrot developed a new financial theory involving fractals that has provided more accurate predictions of market changes. In fact, some of the simulated market lines resembling fractals have been shown to be strikingly similar to real-life market lines.

The process of creating the fractal begins with a price trend line on a price versus time graph. Note that the price and time intervals can be chosen arbitrarily. Once the trend line is set, a broken line called a generator, which corresponds to the real rise and fall oscillations of price, is inserted or substituted into every line segment. Note that a generator with less than three line segments would not represent the rise-and-fall fluctuations of the price in the real market.

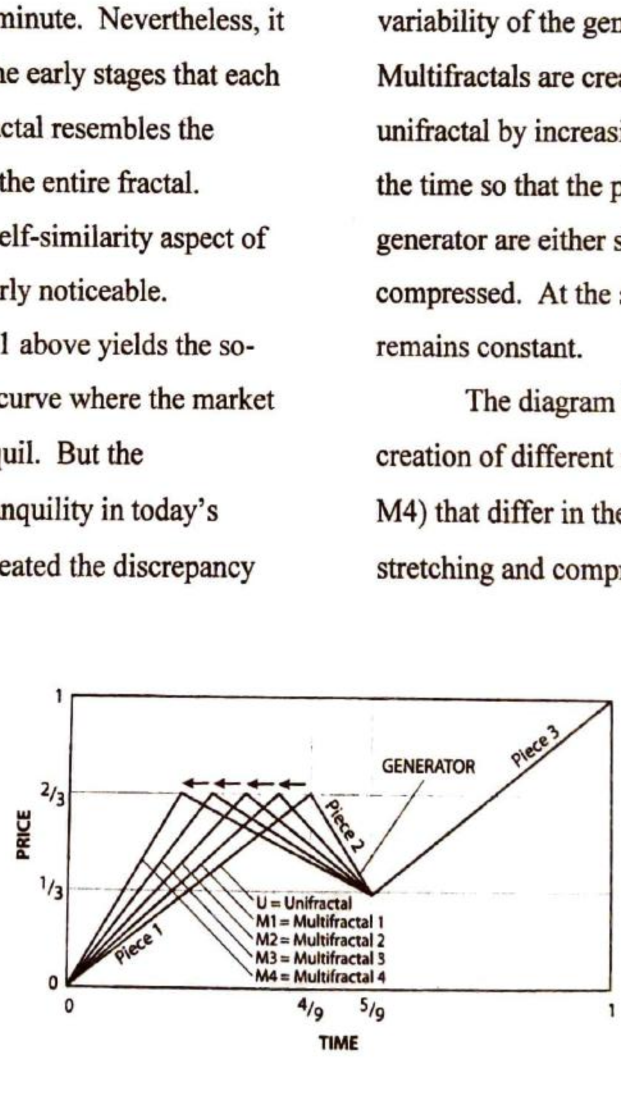
Repeating the interpolations or insertions of the generators reproduces the shape of the generator. The interpolation is a combination of generator iteration and IFS because every part of the base (generator) is substituted with the motif and an identical generator is constantly being substituted into the original generator. The diagrams below illustrate the processes undertaken to achieve the fractal.



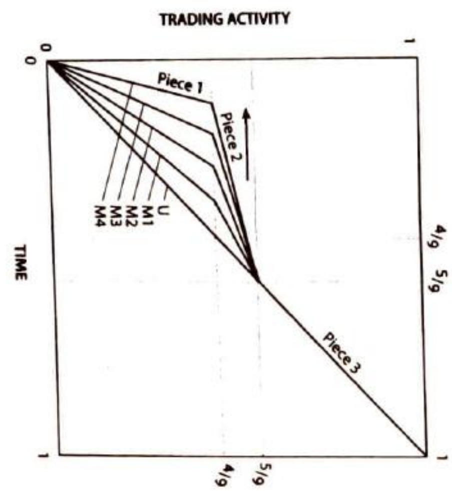
The diagrams only show the early stages of the fractal but iteration can continue infinitely. However, it would make no sense to iterate or interpolate down to negligible time intervals because trading transactions occur in about a minute. Nevertheless, it is evident from the early stages that each "piece" of the fractal resembles the general shape of the entire fractal. Simply put, the self-similarity aspect of the fractal is clearly noticeable.

Diagram 1 above yields the so-called unifractal curve where the market is relatively tranquil. But the assumption of tranquility in today's market is what created the discrepancy for the Portfolio Theory. Therefore, the generator is altered to best simulate the turbulent market periods in addition to the tranquil ones. In essence, the fractal must depict volatility. This is accomplished by introducing the variability of the generators. Multifractals are created from the unifractal by increasing or decreasing the time so that the pieces of the generator are either stretched or compressed. At the same time, the price remains constant.

The diagram below describes the creation of different multifractals (M1 to M4) that differ in the amount of stretching and compressing.



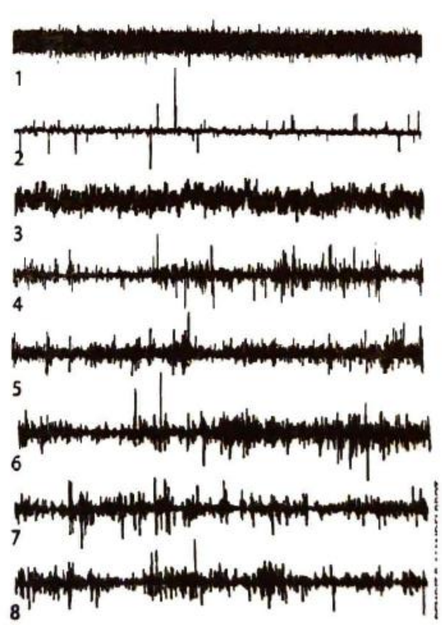
Moving from M1 to M4, the same amount of market activity occurs but in decreasing time intervals. Therefore, the movement on the generator to the left causes market activity to become increasingly volatile. Notice how Piece 1 of the generator becomes shorter while Piece 2 becomes larger. The diagram below shows the increasing volatility caused by the changes in the lengths of Pieces 1 and 2.



Now real price charts can be created from the given data. The price charts shown below have the corresponding alterations of the generator beginning with the original unifractal. Each time Piece 1 is shortened, it produces a price chart that increasingly resembles the characteristics of volatile markets.

Movement of the generator to the left causes market activity to become increasingly volatile

Now to test the validity of this new market theory, eight price charts are shown below to compare actual price charts to simulated multifractals.



Out of the eight price charts, two are real price records while the other six are artificial models. In Charts 1 and 2, there is very little volatility in the price, which labels the charts as unrealistic. Chart 3 has higher amplitude and involves larger yet consistent changes in price. So Chart 3 is also an artificial model due to the absence of an abrupt shift in price. Of the remaining five charts, two happen to be real price records. But now it seems very difficult to differentiate between the artificial and real charts. In fact, Chart 5 refers to the changes in price of the IBM stock and Chart 6 shows the price fluctuations for the dollar-deutsche mark exchange rate. Charts 4, 7, and 8 are completely artificial but notice how they bear a strong resemblance to Charts 5 and 6.

On a practical level, these multifractals cannot predict correct price changes but they provide investors and analysts a helpful estimate of probable market shifts

There are many ways of defining fractals. An informal definition of fractals is processes or images that exhibit self-similarity. Self-similarity exists when reduced versions of the image look similar to the original image.

For example, the fern above is an example of a fractal image. If one were to detach a branch of the fern, it would look very similar to the whole fern itself. This property of self-similarity along with the properties of infinite detail and fractional dimension exists in all fractals.

Mathematics comes into play when generating fractals. One generates fractals by iteration, which is to repeat a certain series of steps that creates a pattern. There are three types of iterations- iterated function system (IFS) iteration, generator iteration, and formula iteration.

Formula iteration is the simplest type of iteration but creates the most complex fractals. For example, the Mandelbrot and the Julia Sets are generated by formula iteration. They are based on taking a number and repeatedly substituting it with another using a mathematical formula or function.

When one views a fractal, he/she should view it as a function on a plane made up of many, many points, or pixels. Each pixel has a x-coordinate and a y-coordinate, which determine its position on the plane. Since each pixel is in a different place, each pixel's coordinates are different from the rest. So to start, a pixel is selected and a function is iterated on the point. This brings us to a new x-coordinate and a new y-coordinate. To iterate a function means to keep applying it over and over, so that is one is to do. One takes the new coordinates, and uses the function again. This brings him/her to a new set of coordinates for that point.

One continues to iterate many, many times until a pattern, the fractal, is created. The type of function determines the type of fractal. If one function is used, it may produce the Mandelbrot set. If another is used, a Julia set might be yielded. There are infinite number of functions, and therefore there are an infinite number of fractals.

An application of fractals is its utilization in the financial markets. A fractal developed by Mandelbrot helps investors and analysts make more accurate predictions. Until recently, investors have adopted the Portfolio Theory, which states that the possibility of a great price change of a stock is not likely. But because major price fluctuations occur at a substantial rate, the Portfolio Theory does not enable an investor to make precise predictions. However with the new financial theory dealing with fractals, more accurate and consistent predictions can be made.

The Witch of Agnesi

By Cindy Lin

Have you ever heard of the Witch of Agnesi? No, it is not a mysterious character that haunts innocent unsuspecting people like that of the Blair Witch Project. It is merely the name given to a mathematical curve, named after Maria Gaetana Agnesi.

Maria Gaetana Agnesi was born in Milan on May 16, 1718, to a wealthy family and she was the oldest of twenty-one children. The fact that her father was a mathematics professor allowed her the educational opportunities that greatly influence her work later on in life. Maria Agnesi was recognized as a child prodigy due to her fluency in Latin, Greek, Hebrew, and several modern languages. Her home was a gathering place for some of the most distinguished intellectuals of the day and Maria Agnesi frequently participated in philosophical and mathematical discussions with them. Despite her high intelligence, Maria was very shy in nature and after her mother's death she took over the management of the household, which she used as an excuse to evade the intellectual discussion seminars with her father's guests. Moreover, it was probably because of the heavy housework load that Maria never married.

By the age of twenty, Maria Agnesi had begun work on her most important work, Analytical Institutions. When it was published in 1748 it caused a sensation in the world of mathematics because it was one of the first and most complete works on finite and infinitesimal analysis. Maria Agnesi is best known for her work on the curve called the "Witch of Agnesi." The curve was called a versiera, a Latin word meaning "to turn." But versiera was also an abbreviation for the Italian word avversiera, meaning the wife of the devil. Thus when Maria Agnesi's work was translated into English the word versiera was confused and translated to the word "witch." The curve thus came to be known as the *Witch of Agnesi*. The curve had previously been studied by Fermat and Guido Grandi in 1703.

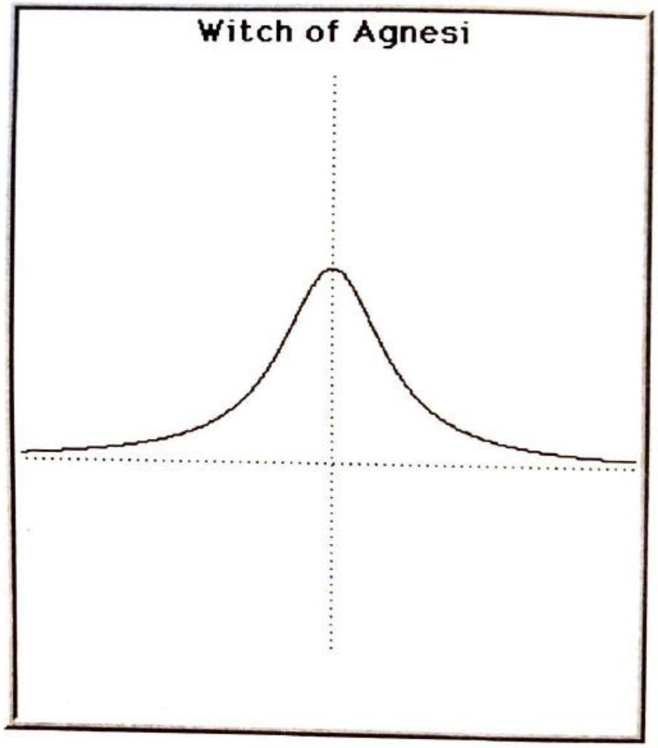
The curve is a versed sine curve and the equation of the curve is

$$y = \frac{8a^3}{x^2 + 4a^2}.$$

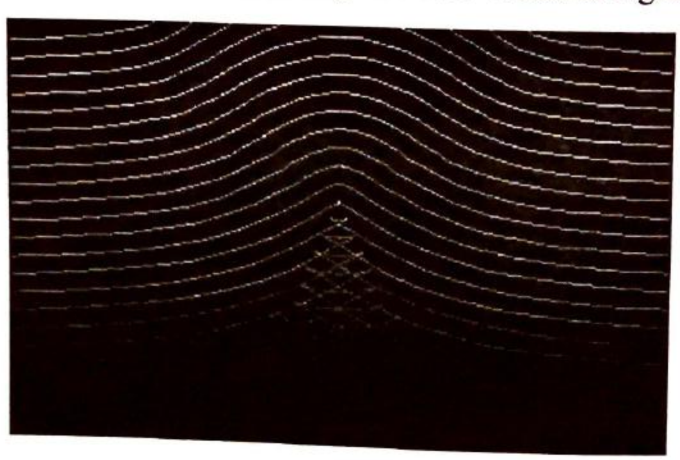
OR

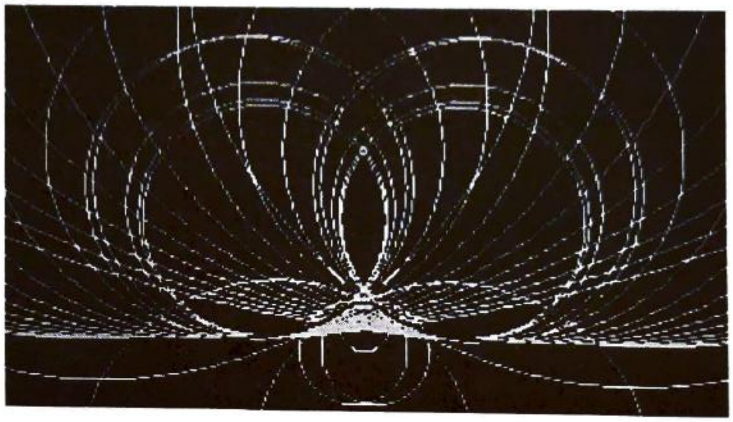
$$X = 2at$$
$$Y = 2a / 1 + t^2$$

This is the basic structure of the Witch of Agnesi:



Below are some more examples of the Witch of Agnesi:





Maria Agnesi's only mathematical inspiration and motivation seemed to come from her father. This is evidenced by the fact that she abandoned her mathematical pursuits and studies after her father's death. She devoted most of her later life to the church and to helping the needy. Maria Agnesi died in 1799 at the age of 81

Chaos and Fractals

Anand Omprakesh

Fractals have been a major force in both the artistic and mathematical world for centuries. Artists have used them for their symmetrical and aesthetic qualities, as well as for their beautiful patterns. Mathematicians have used fractals to illustrate their properties of being sub-divisible and self-similar, or in other words, being infinitely related to themselves.

Chaos is only a phenomenon of the 20^{th} century, but nonetheless, has emerged as an integral part of present day mathematics. A chaotic system appears to be unpredictable, but is really determined by precise deterministic and dynamical laws, just like fractals. Chaotic systems show major fluctuations for small changes in their initial conditions. This can be illustrated in what is known as the butterfly effect, where a wind produced by a butterfly in China could very well provide the dynamics to start a hurricane in the Caribbean.

Chaos and Fractals are intertwined in a number of ways. Both topics have an important role in the current development of mathematics. Most importantly, both play a major function in the growth, development and organization of complex dynamic systems.

Before we can delve into the world of dynamical mathematics, we need to define and clarify certain relevant topics. First of all, a fractal is a certain set X that does not have a clear dimension. Fractal dimensions are fractional, hence fractal, and oftentimes irrational. The essential property of a fractal is that it is self-similar, or in other words, it can be divided into pieces each representing the whole.

It is also important to define the term iteration. When making fractals, a set that has been created must be reiterated into the original function to get a new design. In order for the new design to be self-similar to the whole object, it must create a nearly identical figure.

Now, the essential root of the word fractal is obviously fraction. Why is this so? Basically, a fractal is not merely a self-similar object. Rather, after infinite iterations of the initial self-similar function one begins to see an illustration, but it is an illustration with an obscure dimension. Fractals do not have integral dimensions, but rather fractional ones.

The basics of chaos rely on iteration. The $f^n(x)$ value of x is equal to x substituted into the equation for n iterations.

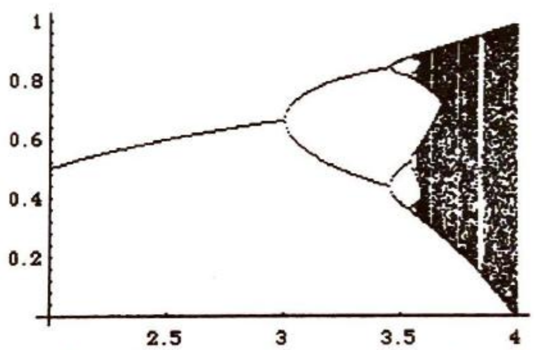
Probably the most common, yet simple, example of chaos theory is the incredible logistic equation:

$$x_2 = ax_1(1-x_1)$$

Where x1 is the initial state, between 1 and 0, and a is the rate of growth. After repeated iterations between a = (0,3) the function stabilizes, or converges upon a single point. However, at a = 3.2, the function bifurcates into two branches, fluctuating between two different points at constant iteration. The function bifurcates once again at a = 3.31, and does this at very small intervals. A bifurcation occurs when a function essentially splits into two different values upon iteration, at least in this case, and at a certain point reaches instability because of an infinite number of branches. This occurs at a = 3.86. At this point the function has transformed into a chaotic function, because the iterations can produce completely random numbers between 0 and 1.

This process of iteration is shown in the diagram graph shown above, where at certain points, such as a = 3.31, the graph bifurcates, or splits into two graphs. This happens constantly, filling up a large portion of the space. However, when we reach a=3.86, the function has reached its stability threshold, and becomes completely chaotic.

We refer to such a graph as a period doubling cascade, because every so often its period number splits. There exists a ratio of x1/x2 = x2/x3 such that at such values the function bifurcates. This ratio is called the Feigenbaum, and is equal to approximately 4.61.



The Lorenz Attractor

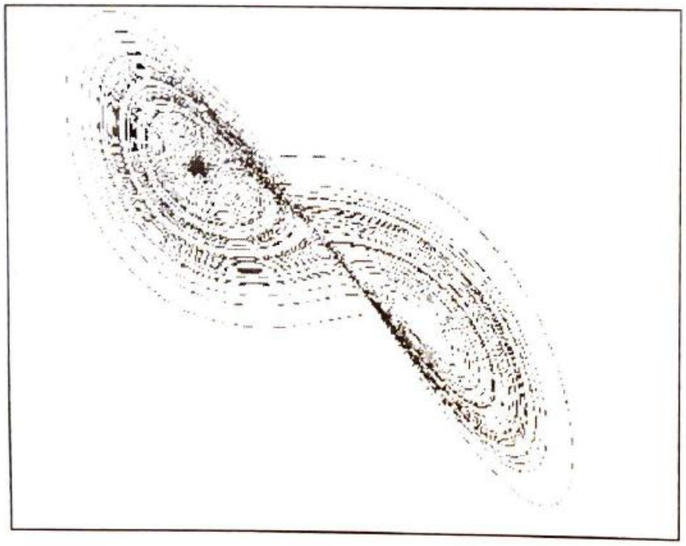
[i]The Lorenz Attractor is an excellent and common example of attractors. In fractals, an attractor is an area that pulls points towards itself, much like the gravity around a star. Those points will remain chaotic and change orbits unpredictably. However, regardless of how chaotic the points become, they will always remain in an orbit around the attractor.

Edward Lorenz discovered this attractor during his research into weather patterns. He did this during the 1960's, before chaos' mainstream acceptance, when the accurate prediction of weather was thought to be easy. While attempting to simulate weather trends in a three dimensional computer model, Lorenz accidentally created this fascinating attractor.

The Lorenz attractor is created by iterating equations, just like many other fractals. It is unique, though, in that it uses three equations, X, Y, and Z, to make a three-dimensional image, not the two dimensional ones that are more common. The three equations used to create this fractal are:

$$x_n = x_{n-1} + d * a * (y_{n-1} - x_{n-1})$$
$$y_n = y_{n-1} + d * (x_{n-1} * (c - z_{n-1}) - y_{n-1})$$
$$z_n = z_{n-1} + d * (x_{n-1} * y_{n-1} - b * z_{n-1})$$

After plotting these equations in a three-dimensional map we get a picture that looks like this:



The Lorenz attractor has two points toward which the rest of the graph appears to converge upon. In other words, they are attractors. The motion around such points is chaotic, but it gradually moves closer to the points.

Lyapunov exponents

Lyapunov exponents are an attempt to convey just how dramatically an error can affect an iteration process. We can thus define such an exponent as the ration of the absolute value of the propagated error and the initial error. For some systems, such as linear systems, the propagated error will be directly related to the initial error, and in such situations there is a linear error.

In chaotic systems, there is an exponential propagated error, and because of this, it is very difficult to make predictions with near-accurate numbers. Thus, it is our goal to determine a single formula that will propagate errors systematically.

We begin by finding a function whose derivative is equal to itself. This function is e^x. From other mathematics you may realize that such a function can be used to analyze the exponential growth of non-linear systems. It is this ability that we use it for this purpose.

Now we take two variables, x and (x + E). By iteration of such variables into a function, we get $f^n(x)$ and $f^n(x + E)$.

Now we can assume that there is some exponential rate of growth, that we call h that $f^n(w+E) - f^n(x) = Ee^{\iota n}$

We solve this equation by using the natural logarithm, which eventually gives us

$$\lambda = \lim_{\substack{t \to \infty \\ |\Delta x_0| \to 0}} \frac{1}{t} \ln \frac{|\Delta x(X_0, t)|}{|\Delta x_0|}$$

where ι is the exponential rate at which the two trajectories of the same function diverge. Dx represents the initial error, which we had discussed as E. t is the time.

The most essential concept here is that we can only get a linear amount of improvement even if we gain exponential precision. For example, in a chaotic system, an initial improvement by tenfold may only improve the result by one.

We can say that if the lyapunov exponent is greater than one, the orbits of two points relatively close to one another initially will separate exponentially by the number h

Lyapunov exponents come in pairs. There is a positive Lyapunov that indicates the exponential rate of growth apart between two trajectories. There is also a negative Lyapunov exponent that indicates their proximity as they come closer to their initial values on a trajectory.

We have defined fractals and chaos and given general examples of these two areas of mathematics. Fractals are essentially images of a fractional dimension. Chaotic functions tend to produce seemingly random results from, oftentimes, apparently harmless equations through the use of the iterative process. We also examined certain chaotic functions' phase space, and were even able to determine that some possessed fractal dimensions. Thus, in our efforts to generally examine chaotic functions and aspects of fractals, we are able to integrate them in a very subtle, yet significant way.

A Proof of Euler's Equation using Graph Theory and Map Coloring

Abhra Haldar

The beauty of mathematics lies in its universal applicability. The great challenges of the field are often those with the most elegant and practical application. Difficult mathematical problems are often left unsolved for centuries until the combined insights of many great minds, both past and present, tackle the puzzle. Once solved, however, these problems lend a wealth of applications and discovery to society.

The field of graph theory is perhaps the most obviously applicable branch of pure mathematics. The use of points and lines to represent actual quantities in space has unlimited applications in the real world. One of the more obscure uses of graph theory is the coloring of maps.

When one thinks of a map, one immediately envisions a simple multicolored board filled with the names of different geographic regions. We have all colored as children and found it a pleasant activity. Yet a true challenge lies in coloring a map so that no two adjacent regions have the same color. Although it has been proven that four colors are sufficient to color any map, the process of map coloring is greatly simplified by the use of graph theory.

If we let each region of the map be represented by a point (vertex), and all adjacent regions be connected by a line (edge), we are left with a closed, connect-the-dot shape known as the *dual graph* of a map. An edge is *incident* to a vertex if it connects the point to another vertex. The *degree* of a vertex is the number of vertices incident to it. A *connected graph* is one in which every vertex is connected to at least on other. An important notion is that of a *bridge*, an edge in a connected graph that, if removed, would make it disconnected. A graph whose edges are all bridges is known as a *tree.* (Note that the terms graph and map are use interchangeably from here on)

One of the most important tools in analyzing graphs (and thus maps) is Euler's equation. This formula relates the number of vertices (V) and edges (E) of the dual graph of a map with C number of countries. Euler's equation states that for any planar graph, $V - E + C = 2$ (where the unbounded plane surrounding the graph is regarded as a region). The proof of this theorem using normal geometric techniques is rather lengthy and difficult, but is much more elegant using graph theory.

To prove this useful formula, one can use mathematical induction on number of edges E. Let M be a map with V vertices, E edges, and C countries, with D as its dual graph. If $E = O$, then $C = 1$ (the unbounded plane) and $V = 1$ (since a dual graph must have at least one point). Thus, $V - E + C = 2$. It is assumed that the theorem holds for E. A proof must be formulated for $E + 1$. One of two conditions must exist: either the graph has an edge that is not a bridge or it does not. However, a connected graph that has edges that are all bridges is known as a tree. Thus, the latter of the two conditions describes a tree. In a tree, $C = 1$ (the unbounded plane). By another theorem, if one has a tree with n vertices and m edges, then $n - m = 1$ (proved simply by induction). Then, $V - E = 1$ and $V - E + C = 2$.

If the graph with V vertices, E edges, and C countries has an edge d which is not a bridge, let D* be the connected graph obtained by removing d and let M* be the corresponding map. Now, D* has E - 1 edges, and from the induction hypothesis, V - (E-1) + C*=2, where C* is the number of regions in M*. Because the edge d is not a bridge, it must separate two distinct regions, so if it was removed, then there would C-1 regions remaining. Thus, C¢ = C - 1. By substitution into the previously derived equation:

V - (E -1) + C - 1 = 2

V - E + 1 -1 + C = **2 = V - E +C**

Thus, Euler's equation has been proved using graph theory and map coloring techniques.

The quantity V – E + C becomes essential in analyzing the properties of more complex, multi-dimensional shapes (and does not necessarily equal 2). By properly adapting the definition of a graph to fit the scenario, these cases can be proven similarly as well. In conclusion, the proof of Euler's theorem using map coloring and graph theory is one of more elegant confirmations of the result and evidence of the interdependence and applicability of mathematics.

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